Ziquan Wei

+1-919-923-8297 | ziquanw@email.unc.edu | ziquanw.com | Linkedin | GitHub | 640 citations

Education

- **University of North Carolina at Chapel Hill, USA** — ***PhD in Computer Science***, *Aug 2023 - May 2028*
- **Huazhong University of Science and Technology, China** — ***Master in Biomedical Engineering***, *Sept 2019 - June 2022*
- **Yunnan Normal University, China** — ***Bachelor in Computer Science***, *Sept 2015 - June 2019*

Research Interest

AI for healthcare: personalized, individual-level models that turn routine and consumer-grade signals (EHR, imaging, ECG / EEG) into actionable predictions. I also work on the deployment questions that decide whether such models help patients: cross-site robustness, label scarcity, and on-device privacy.

Research Experience

- **University of North Carolina at Chapel Hill [🌐]** — ***Research Assistant***, *Sept 2022 - Current*
  - **Personalized Digital Twins**: Built DiffDT, an SDoH-conditioned latent diffusion model linking tokenized healthcare events to multi-organ sensor data on UK Biobank (∼500K medical histories; 44.8K brain / 24.0K heart / 28.7K liver / 32.2K kidney scans), enabling in-silico intervention and per-patient disease-trajectory forecasting [3]; and Time2Track, generating individualized 4D cardiac motion conditioned on electrophysiology—a cardiac digital twin driven by ECG [5].
  - **Clinical Foundation Models**: Pretrained the Large Connectome Model (1.2B) on brain–environment interactions and BrainMoE, a Mixture-of-Experts over cognitive states, across UK Biobank / HCP / ADNI / OASIS / ABIDE / PPMI for early diagnosis of Autism, Parkinson's, Alzheimer's, and Schizophrenia; added BECA meta-matching for zero-shot transfer to unseen disease cohorts where labels are scarce [1][14][13].
  - **Trustworthy Deployment**: Studied replicability of deep models across sites, scanners, and pipelines [7][9]; administer a 10-node, 20-GPU, 1.5PB cluster.
- **EGRA.ai [🌐]** — ***Research Intern***, *May 2026 – Aug 2026*
  - **RL from Biofeedback**: Extended RLHF with a reward model driven by neural biofeedback, aligning LLMs to consumer-grade physiological signals rather than preference labels alone.
- **Huazhong University of Science and Technology** — ***Graduate Researcher (MSc)***, *Sept 2019 – June 2022*
  - **Cancer Screening at Scale**: First-authored YOLCO, cutting a 4-gigapixel cytology slide to 55.9s inference [27], and PolarNet for out-of-distribution glandular-cell detection [26]; contributed to the multi-center cervical cancer screening system in Nature Communications (3,545 slides, 5 hospitals, 5 scanners; 95.1% sensitivity / 93.5% specificity on 1,170 independent slides) [29].

Projects

- **CyberNeuro: A Privacy-Preserving, On-Device Medical AI Assistant** — *Jan 2026 - May 2026*
  Tools: [MedGemma, Ollama, FastMCP, Agent-to-Agent, FastAPI, TypeScript, React] — [GitHub, Kaggle Writeup, Demo]
  - **Led the project** (47 of 84 commits, 6-person team) for the Kaggle MedGemma Impact Challenge: users chat with their own biomarker tables and MRI while every inference runs on-device via Ollama (MedGemma 1.5 4B), so PHI never leaves the machine.
  - **Built an 11-role agent system** (planner → validator → executor → researcher → reporter) over an MCP tool layer that grounds every answer in PubMed / OpenAlex evidence and normative reference curves, with bounded self-correction and state rollback; on 500 clinical queries it reaches 93.6% end-to-end task success and 95.8% routing accuracy (100% precision).

Skills

- **Personalized & Clinical AI:** Conditional diffusion & digital twins, longitudinal disease-progression modeling, foundation-model pretraining (1.2B), Mixture-of-Experts, zero-shot meta-matching under label scarcity, multi-site robustness, early diagnosis & risk stratification, RLHF, LLM agents with MCP tool use
- **Health Data:** UK Biobank, HCP, ADNI, OASIS, ABIDE, PPMI; EHR / ICD event sequences, social determinants of health; fMRI, DWI, PET, T1, ECG, EEG, gigapixel pathology WSI, teravoxel LSFM; BIDS, DICOM, NIfTI, on-device inference
- **Engineering:** Python, PyTorch, C++, TypeScript / React, FastAPI, MedGemma / Gemma / Qwen, Ollama, FastMCP, Nextflow, multi-GPU distributed training, HPC administration (10 nodes, 20 GPUs, 1.5PB)

PUBLICATIONS

†=EQUAL CONTRIBUTION

[1] Wei Z., Dan T., Wu G. **Large Connectome Model: An fMRI Foundation Model of Brain Connectomes Empowered by Brain-Environment Interaction in Multitask Learning Landscape**. *AAAI '26*.

[2] Wei Z., Wu G. **Teravoxel Microscopy Image Analysis for Neurological Diseases**. *ARBME '26* (IF=11.5).

[3] Wei Z., Dan T., Wu G. **Marrying Generative Model of Healthcare Events with Digital Twin of Social Determinants of Health for Disease Reasoning**. *ICML '26*.

[4] Wei Z., Dan T., Ding J., Laurienti P.J., Wu G. **NeuroDetour: A Neural Pathway Transformer for Uncovering Structural-Functional Coupling Mechanisms in Human Connectome**. *MedIA '26* (IF=14.0).

[5] Wei Z., et al. **From Heartbeat to Cardiochoreography: A Mechanics Foundation Model for Individualized 4D Cardiac Motion Generation Conditioned on Electrophysiology**. *NeurIPS '26* (In submission, ratings 5 4 4).

[6] Wei Z., Curtin I., Kyere F.A., et al. **CellPheno: A High-throughput Computational Platform for Quantifying Cellular Resolution Whole Brain Microscopy Images**. *bioRxiv '26*.

[7] Ding J., Dan T., Wei Z., Laurienti P.J., Wu G. **Scanning the Horizon of Replicability in Neuroscience: A Recipe of Developing Replicable Deep Models for Functional Neuroimages**. *IEEE TBME '26* (IF=4.5).

[8] Wen T., Wei Z., Dan T., et al. **Image Imputation for Real-World Defective Microscopy Images of Mouse Brains**. *ISBI '26*.

[9] Ding J., Dan T., Wei Z., Laurienti P.J., Wu G. **Machine learning on dynamic functional connectivity: Promise, pitfalls, and interpretations**. *Inf. Sci. '26* (IF=6.0).

[10] Chen Y., Li Y., Tong J., . . . , Wei Z., et al. **NeuroPilot: An Agent-Driven Smart Pipeline for Processing, Quality Control, and Managing Neuroimages**. *arXiv '26*.

[11] Kyere F.†, Curtin I.†, Wei Z.†, et al. **Chd8 haploinsufficiency leads to molecular layer heterotopias and age-dependent cortical expansion**. *bioRxiv '26*.

[12] Wei Z., Dan T., Ding J., Wu G. **Efficient Graph Representation Learning by Non-Local Information Exchange**. *Electronics '25* (IF=2.9).

[13] Wei Z., Dan T., Wu G. **Brain-Environment Cross-Attention (BECA) Meta-matching: A New Perspective of Brain Connectome Zero-Shot Learning**. *MICCAI '25* (Early accept (9%)).

[14] Wei Z., Dan T., Chen T., Wu G. **BrainMoE: Cognition Joint Embedding via Mixture-of-Expert Towards Robust Brain Foundation Model**. *NeurIPS '25*.

[15] Cho H., Wei Z., Lee S., et al. **Conditional Diffusion Model using Ordinal Regression for Longitudinal Neurodegenerative Data Generation**. *Alz. & Dem. '25* (IF=12.8).

[16] Cho H., Wei Z., Lee S., et al. **Conditional Diffusion with Ordinal Regression: Longitudinal Data Generation for Neurodegenerative Disease Studies**. *ICLR '25* (Spotlight poster (3.26%)).

[17] Wei Z., Dan T., Ding J., Laurienti P., Wu G. **Representing Functional Connectivity with Structural Detour: A New Perspective to Decipher Structure-Function Coupling Mechanism**. *MICCAI '24*.

[18] Wei Z., Dan T., Ding J., Wu G. **NeuroPath: A Neural Pathway Transformer for Joining the Dots of Human Connectomes**. *NeurIPS '24*.

[19] Wei Z., Wu G. **Non-local Exchange: Introduce Non-locality via Graph Re-wiring to Graph Neural Networks**. *NeurIPS-W '24*.

[20] Dan T., Wei Z., Kim W.H., Wu G. **Exploring the Enigma of Neural Dynamics Through A Scattering-Transform Mixer Landscape for Riemannian Manifold**. *ICML '24*.

[21] Ding J., Dan T., Wei Z., Laurienti P., Wu G. **A Wasserstein Recipe for Replicable Machine Learning on Functional Neuroimages**. *MICCAI '24*.

[22] Wu Y., Liu Y., Li Y., . . . , Wei Z., et al. **Symmetry engineering in 2D bioelectronics facilitating augmented biosensing interfaces**. *PNAS '24* (IF=9.5).

[23] Wei Z., Dan T., Ding J., et al. **High Throughput Deep Model of 3D Nucleus Instance Segmentation by Stereo Stitching Contextual Gaps**. *ISBI '23*.

[24] Wei Z., Dan T., Ding J., Dere M., Wu G. **A General Stitching Solution for Whole-Brain 3D Nuclei Instance Segmentation from Microscopy Images**. *MICCAI '23* (Early accept (14%)).

[25] Dan T., Ding J., Wei Z., et al. **Re-think and re-design graph neural networks in spaces of continuous graph diffusion functionals**. *NeurIPS '23*.

[26] Wei Z., Cheng S., Cai J., et al. **Cervical Glandular Cell Detection from Whole Slide Image with Out-Of-Distribution Data**. *arXiv '22*.

[27] Wei Z., Cheng S., Hu J., et al. **An Efficient Cervical Whole Slide Image Analysis Framework Based on Multi-scale Semantic and Location Deep Features**. *arXiv '21*.

[28] Zhang S.†, Ding Y.†, Wei Z.†, Guan C. **Continuous emotion recognition with audio-visual leader-follower attentive fusion**. *ICCVW '21*.

[29] Cheng S., Liu S., Yu J., . . . , Wei Z., et al. **Robust whole slide image analysis for cervical cancer screening using deep learning**. *Nat. Comm. '21* (IF=18.1).

[30] Yang Z., Yang Y., Yang K., Wei Z.. **Non-rigid image registration with dynamic Gaussian component density and space curvature preservation**. *IEEE TIP '18* (IF=13.7).

[31] Zhang S., Yang K., Yang Y., Luo Y., Wei Z.. **Non-rigid point set registration using dual-feature finite mixture model and global-local structural preservation**. *Pattern Recognit. '18* (IF=9.1).

[32] Wei Z., Han Y., Li M., et al. **A small UAV based multi-temporal image registration for dynamic agricultural terrace monitoring**. *Remote Sens. '17* (IF=4.3).

HONORS & SERVICE

- **Honors:** Editor of Distinction Award 2025, Scientific Reports (editorial board); Graduate School Transportation Grant, UNC-Chapel Hill; Outstanding Graduate, Huazhong University of Science and Technology
- **Conference Reviewer:** NeurIPS, ICML, ICLR, AAAI, AISTATS, MICCAI
- **Journal Reviewer:** IEEE Transactions on Medical Imaging, NeuroImage, Pattern Recognition

TALKS

- **2025 Tissue Clearing & Imaging Conference, UNC-Chapel Hill [🌐]** *Speaker*, *Apr 2025*
  - **CellPheno**: A High-throughput Computational Platform of Cellular Phenotyping for Real-world Whole Brain Microscopy Images.